

Data Management
Prof. Andrea Maurino
Master's degree in Data Science
Academic Year 2023/2024
July 8th 2024

60 minutes

Question 1

Create a database to train a model that estimates train delays. The 'treniInOrario' website provides APIs that, given a train number, show the real-time schedule of when the train passed through various stations and the expected schedule for the subsequent stations. The 'RTI' website provides information about all stations in Italy (name, latitude, longitude, number of tracks, etc.) and the trains that pass from them. Additionally, the 'previsioniMeteo' website shows real-time weather data, but it does not have public APIs.

Define:

- A reasonable data acquisition strategy to obtain most of the aforementioned information, including the historical evolution of data
- The data model used to store the information and the motivations behind it.
- Strategies for integrating data sources.
- Quality dimensions to consider.

Question 2

Data quality is a critical aspect of data management that directly impacts the reliability and effectiveness of data-driven decision-making. Explain the key dimensions of data quality and discuss at least one methods used to ensure high data quality

Question 3

Compare and contrast the different types of NoSQL databases (e.g., document, and graph databases). Discuss the scenarios in which each type would be most appropriate, considering factors such as data structure, scalability, and query requirements.